



RNA editing one-step closer

Scientists at Stalk University mapped molecular structure of CRISPR enzyme, could help precise cell manipulation. <https://digit.in/oct18-76-1>



AR to the rescue

Researchers have developed AR based systems to help less experienced doctors perform complex procedures. <https://digit.in/oct18-76-2>

CRISPR v. Autism

An ethico-scientific debate

Recent CRISPR advances could cure autism. The debate now rages on whether it should.

Arnab Mukherjee | arnab@digit.in

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ince the advent of genome editing technology, the ethics debate hasn't been far behind. At Digit,

we've covered the topics of designer babies, superior humans and more. On one side you have the proponents of the sanctity of human DNA who argue that editing the human genome is equivalent to playing god, something we're not supposed to do. On the other side you have the flag bearers of the healthcare benefits of this technology and the possibility of mitigating and eradicating conditions like cancer, diabetes and perhaps the most debated of them all - autism. And at the very centre of this debate is CRISPR-Cas9 technology, an affordable gene editing methodology which promises to be within the grasp of the common man in our lifetime. Recently, scientists



CRISPR-Cas9 makes gene editing accessible

have made unforeseen advancements in using this technology to reduce autism in mice, putting CRISPR-Cas9 research, and the debate associated with it, into the spotlight yet again.

THE BASICS

While most geeks, even with a passing interest in recent R&D, do know of CRISPR, there's no harm in going over some of the basics pertaining to this article once more. Clustered Regularly Interspaced Short Palindromic Repeats and CRISPR-associated protein 9, a.k.a CRISPR-Cas9, is a gene editing methodology that relies on using a

naturally occurring gene editing capability in bacteria. When a bacteria is infected by a virus or more specifically, viral DNA, its only way to protect itself is to edit the DNA out of itself. It achieves this using the Cas9 protein that binds itself with an RNA copy of the target DNA. This combination then scours the cell to find matching DNA and edit it out, not too different from a scissor, if you will. This also allows for inserting donor DNA at the cut site - but that's not the focus here.

On the other side, autism or autism spectrum disorder refers to a wide range of conditions mainly identified